

HUMAN COMPUTER INTERACTION

UNIT-II

HCI in the software process
Design Rules
Evaluation Techniques
Universal Design

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RMDEC

HCI IN THE SOFTWARE PROCESS

- Software Life Cycle
- Usability engineering
- Iterative design and prototyping
- Design rationale

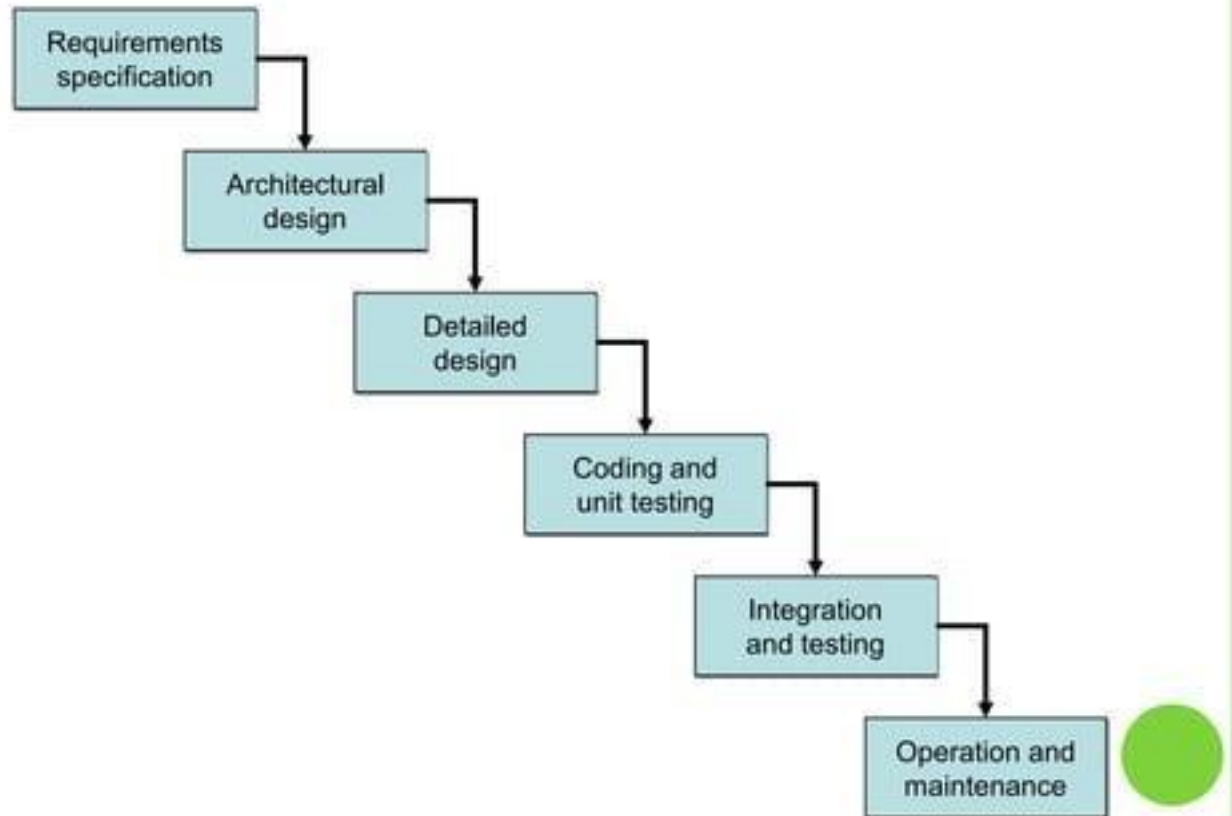


THE SOFTWARE LIFECYCLE

- Software engineering is the discipline for understanding the software design process, or life cycle
- Designing for usability occurs at all stages of the life cycle, not as a single isolated activity



THE WATERFALL MODEL



ACTIVITIES IN THE LIFE CYCLE

Requirements specification

designer and customer try capture what the system is expected to provide can be expressed in natural language or more precise languages, such as a task analysis would provide

Architectural design

high-level description of how the system will provide the services required factor system into major components of the system and how they are interrelated needs to satisfy both functional and nonfunctional requirements

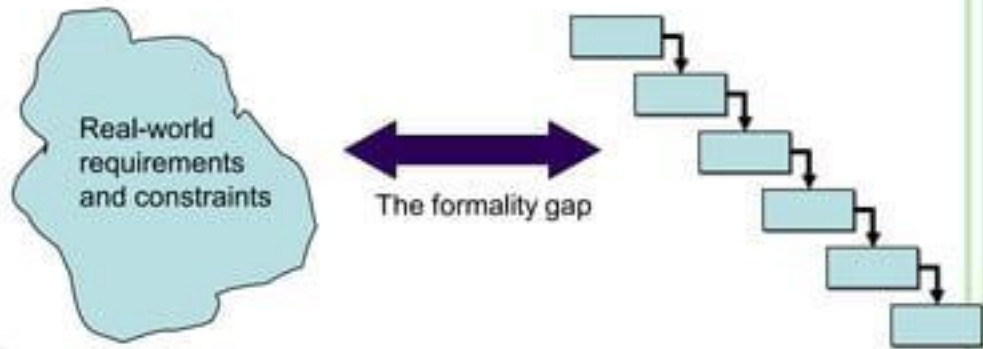
Detailed design

refinement of architectural components and interrelations to identify modules to be implemented separately the refinement is governed by the nonfunctional requirements



VERIFICATION AND VALIDATION

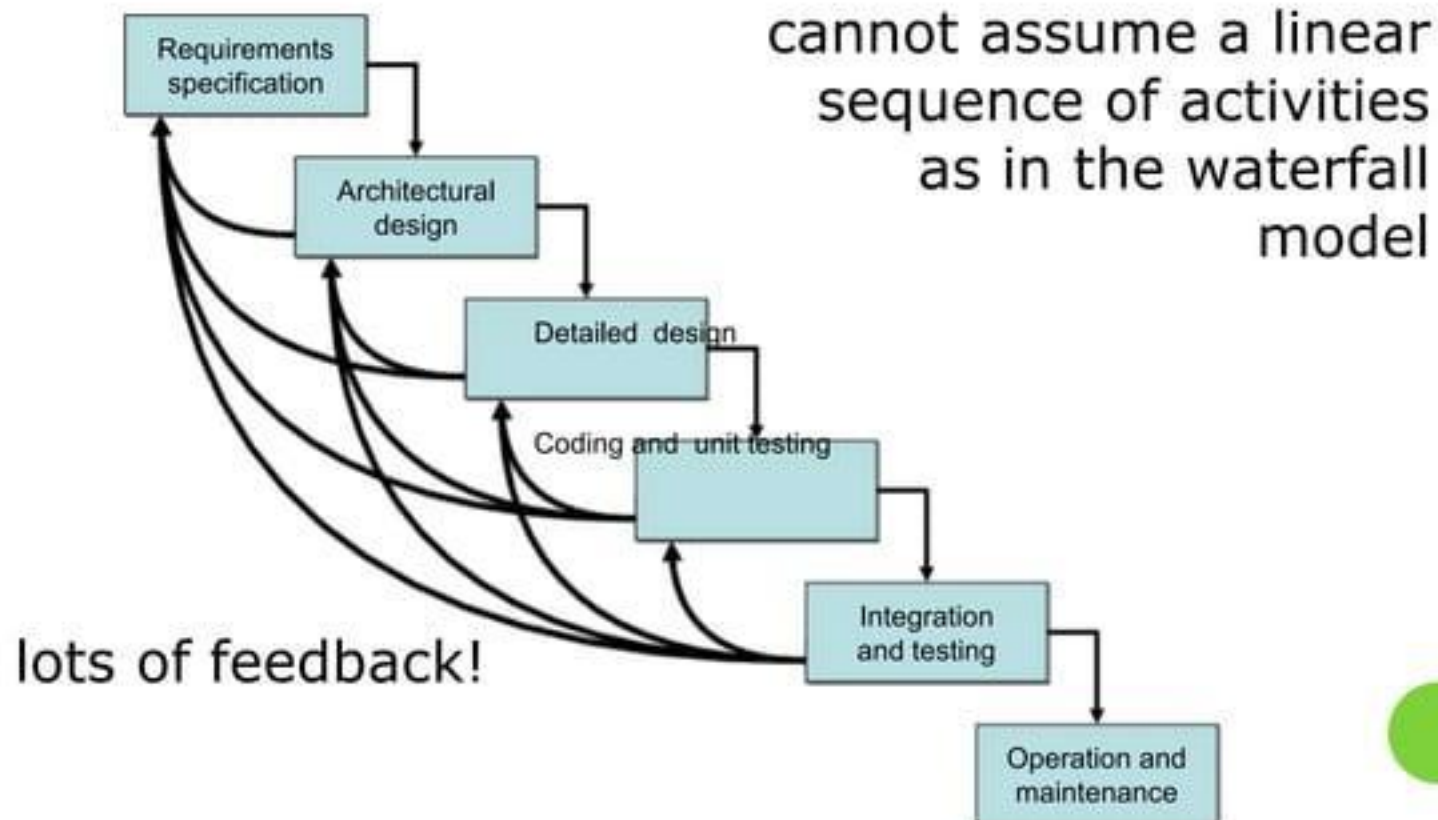
Verification
designing the product right
Validation
designing the right product



The formality gap
validation will always rely to some extent on subjective means
of proof
Management and contractual issues
design in commercial and legal contexts



THE LIFE CYCLE FOR INTERACTIVE SYSTEMS



USABILITY ENGINEERING

What is Usability?

- **Usability** is the degree to which a software can be used by specified consumers to achieve quantified objectives with effectiveness, efficiency, and satisfaction in a quantified context of use.

Usability Engineering

- devising human-computer interfaces that have high usability or user friendliness.
- It provides structured methods for achieving efficiency and elegance in interface design



- Usability engineering is the inclusion of a usability specification, forming part of the requirements specification, that concentrates on features of the user–system interaction which contribute to the usability of the product.
- Various attributes of the system are suggested as gauges for testing the usability.
- For each attribute, six items are defined to form the usability specification of that attribute.
 - measuring concept
 - measuring method
 - now level
 - worst case
 - planned level
 - best case



PART OF A USABILITY SPECIFICATION FOR A VCR

Attribute: Backward recoverability

Measuring concept:	Undo an erroneous programming sequence
Measuring method:	Number of explicit user actions to undo current program
Now level:	No current product allows such an undo
Worst case:	As many actions as it takes to program-in mistake
Planned level:	A maximum of two explicit user actions
Best case:	One explicit cancel action



- **Recoverability** - ability to reach a desired goal after recognition of some error in previous interaction.
- **measuring concept**- attributes in terms of actual product.
- **measuring method** states how the attribute will be measured
- **now level**- *value for the measurement with the existing system, whether it is computer based or not.*
- **worst case value** - *lowest acceptable measurement* for the task, providing a clear distinction between what will be acceptable and what will be unacceptable in the final product.
- **planned level** - *target* for the design
- **best case**- *best possible* measurement given the current state of development tools and technology.



SOME METRICS FROM ISO 9241

- measurement criteria which can be used to determine the measuring method for a usability attribute is called usability metrics.

Usability objective	Effectiveness measures	Efficiency measures	Satisfaction measures
Suitability for the task	Percentage of goals achieved	Time to complete a task	Rating scale for satisfaction
Appropriate for trained users	Number of power features used	Relative efficiency compared with an expert user	Rating scale for satisfaction with power features
Learnability	Percentage of functions learned	Time to learn criterion	Rating scale for ease of learning
Error tolerance	Percentage of errors corrected successfully	Time spent on correcting errors	Rating scale for error handling



ENGINEERING

- rely on measurements of very specific user actions in very specific situations.
- provides a means of satisfying usability specifications and not necessarily usability



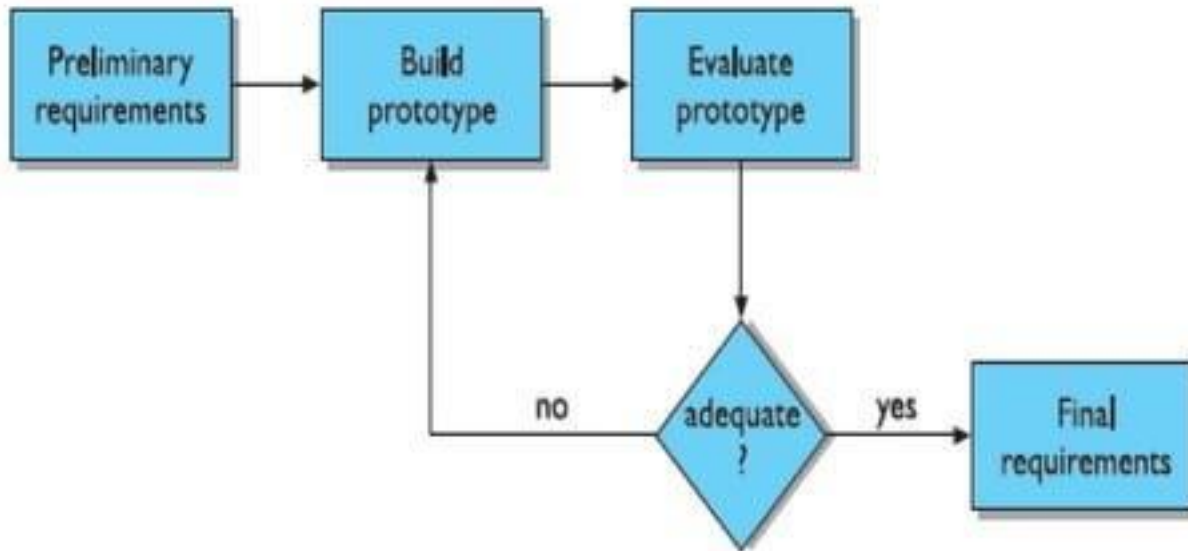
ITERATIVE DESIGN AND PROTOTYPING

- Iterative design tries to overcome the inherent problems of incomplete requirements specification by cycling through several designs, incrementally improving upon the final product with each pass.
- Iterative design is described by the use of *prototypes*.
- 3 different types of prototypes
 - throw-away
 - incremental
 - evolutionary
- Management issues
 - time
 - planning
 - non-functional features
 - contracts



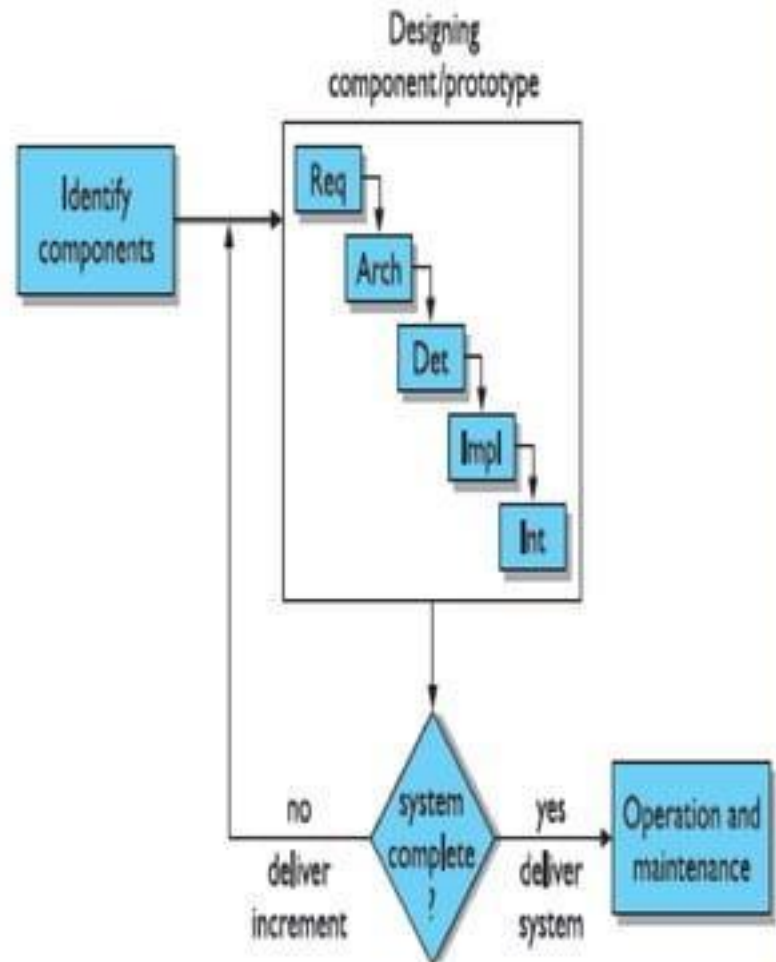
THROW-AWAY

- The prototype is built and tested.
- The design knowledge gained from this exercise is used to build the final product, but the **actual prototype is discarded**



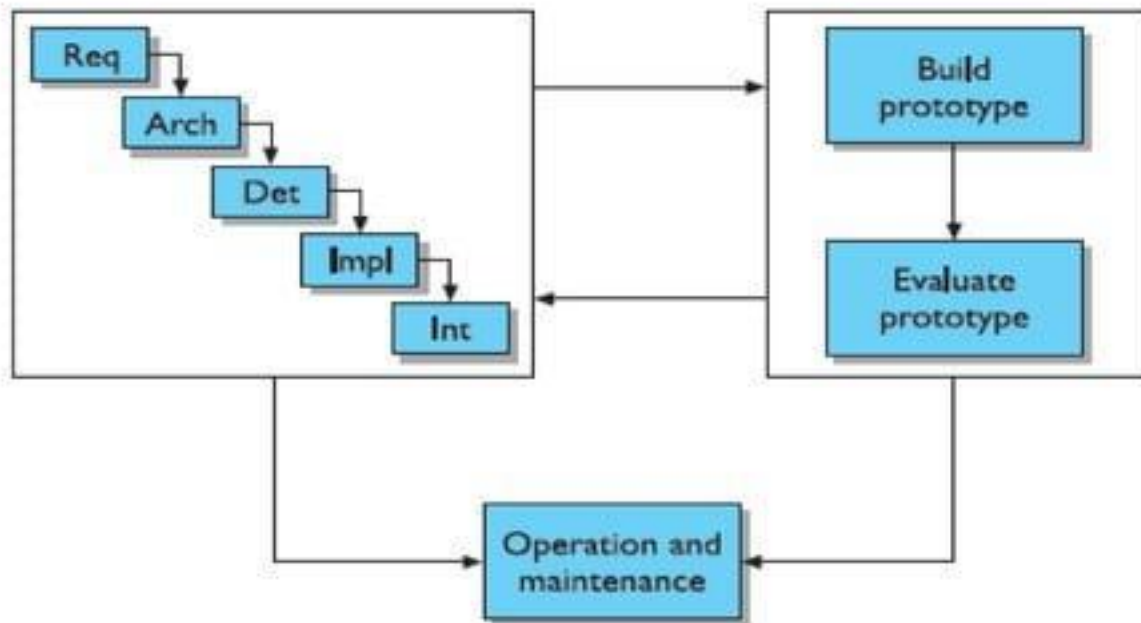
INCREMENTAL

- There is one overall design for the final system, but it is partitioned into independent and smaller components.
- The final product is then released as a series of products, each subsequent release including one more component.



EVOLUTIONARY

- Here the prototype is not discarded and serves as the basis for the next iteration of design.
- The actual system is seen as evolving from a very limited initial version to its final release



TECHNIQUES FOR PROTOTYPING

Storyboards

- Graphical depiction of the outward appearance of the intended system, without any accompanying system functionality.
- The origins of storyboards are in the film industry, where a series of panels roughly depicts snapshots from an intended film sequence in order to get the idea across about the eventual scene.

Limited functionality simulations

- designer can rapidly build graphical and textual interaction objects and attach some behavior to those objects, which mimics the system's functionality.

- plenty of prototyping tools available for the rapid development of simulation prototypes.
- They provide a quick development process for a very wide range of small but highly interactive applications.
- Eg:
 1. *HyperCard*, a simulation environment for the Macintosh line of Apple computers. Simulations produced are throw-away prototypes
 2. *Wizard of Oz technique* - does not require very much computer supported functionality .Designers can develop a limited functionality prototype and enhance its functionality in evaluation by providing the missing functionality through human intervention.



High-level programming support

- o HyperTalk - attach functional behavior to the specific interactions that the user will be able to do, such as position and click on the mouse over a button on the screen
- o *user interface management system /UIMS* - to connect the behavior at the interface with the underlying functionality.



DESIGN RATIONALE

Design rationale is information that explains why a computer system is the way it is.

- including its structural or architectural description and its functional or behavioral description.

Benefits of design rationale

- communication throughout life cycle
- reuse of design knowledge across products
- enforces design discipline
- presents arguments for design trade-offs
- organizes potentially large design space
- capturing contextual information



DESIGN RATIONALE (CONT'D)

Types of DR:

- Process-oriented
 - preserves order of deliberation and decision-making
- Structure-oriented
 - emphasizes post hoc structuring of considered design alternatives
- Two examples:
 - Issue-based information system (IBIS)
 - Design space analysis

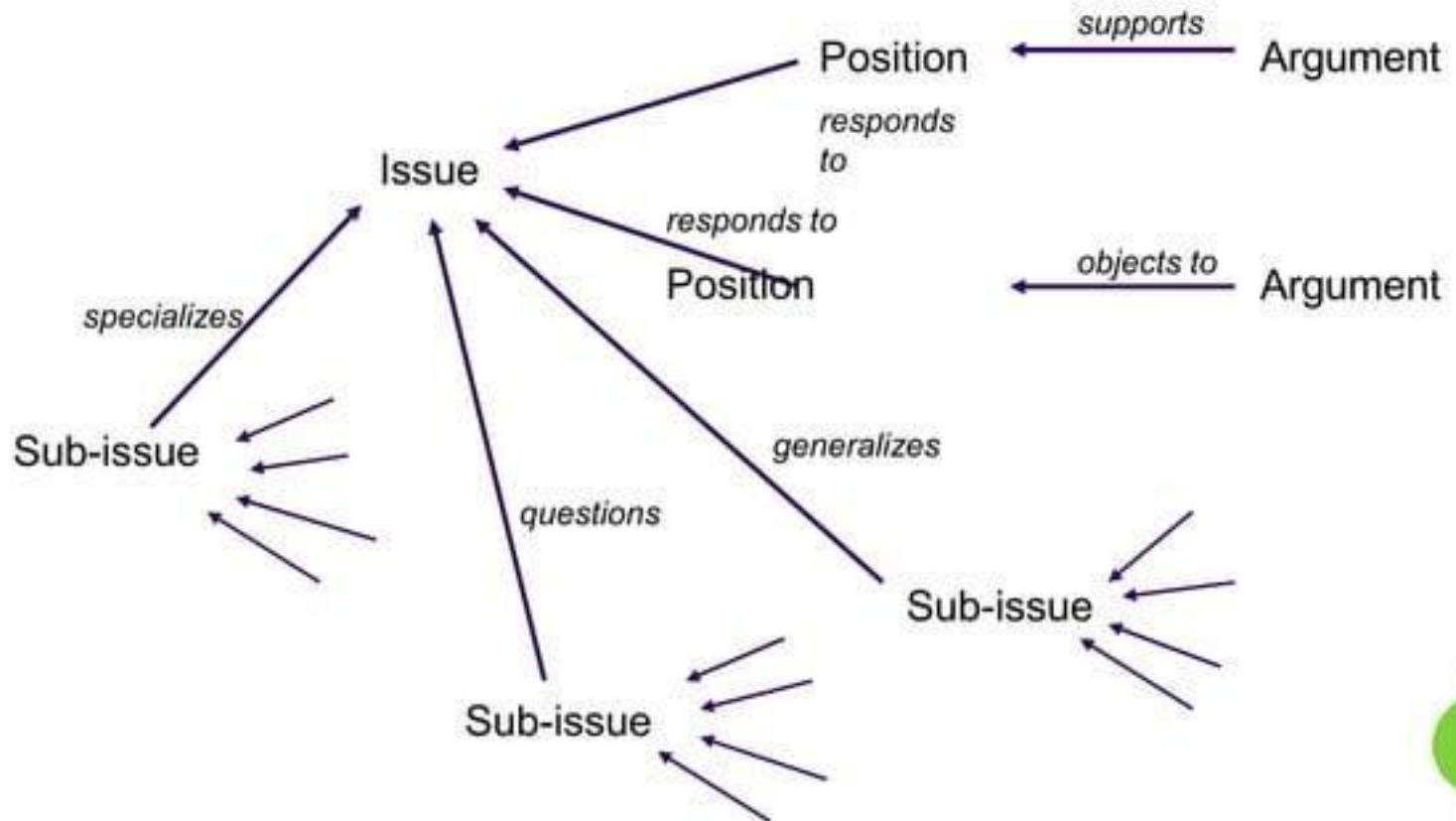


Process-oriented DR

ISSUE-BASED INFORMATION SYSTEM (IBIS)

- style for representing design and planning dialog
 - main elements:
 1. Root issues - represents the main problem or question that the argument is addressing
 2. Positions - potential resolutions of an issue
 3. Arguments - modify the relationship between positions and issues
 - hierarchy grows as secondary issues are raised which modify the root issue in some way.
 - secondary issues are in turn expanded by positions and arguments, further sub-issues, and so on.
 - gIBIS is a graphical version
- 

STRUCTURE OF GIBIS



DESIGN SPACE ANALYSIS

- Design space is initially structured by a set of questions representing the major issues of the design
- QOC – hierarchical structure:

questions (and sub-questions)

- represent major issues of a design

options

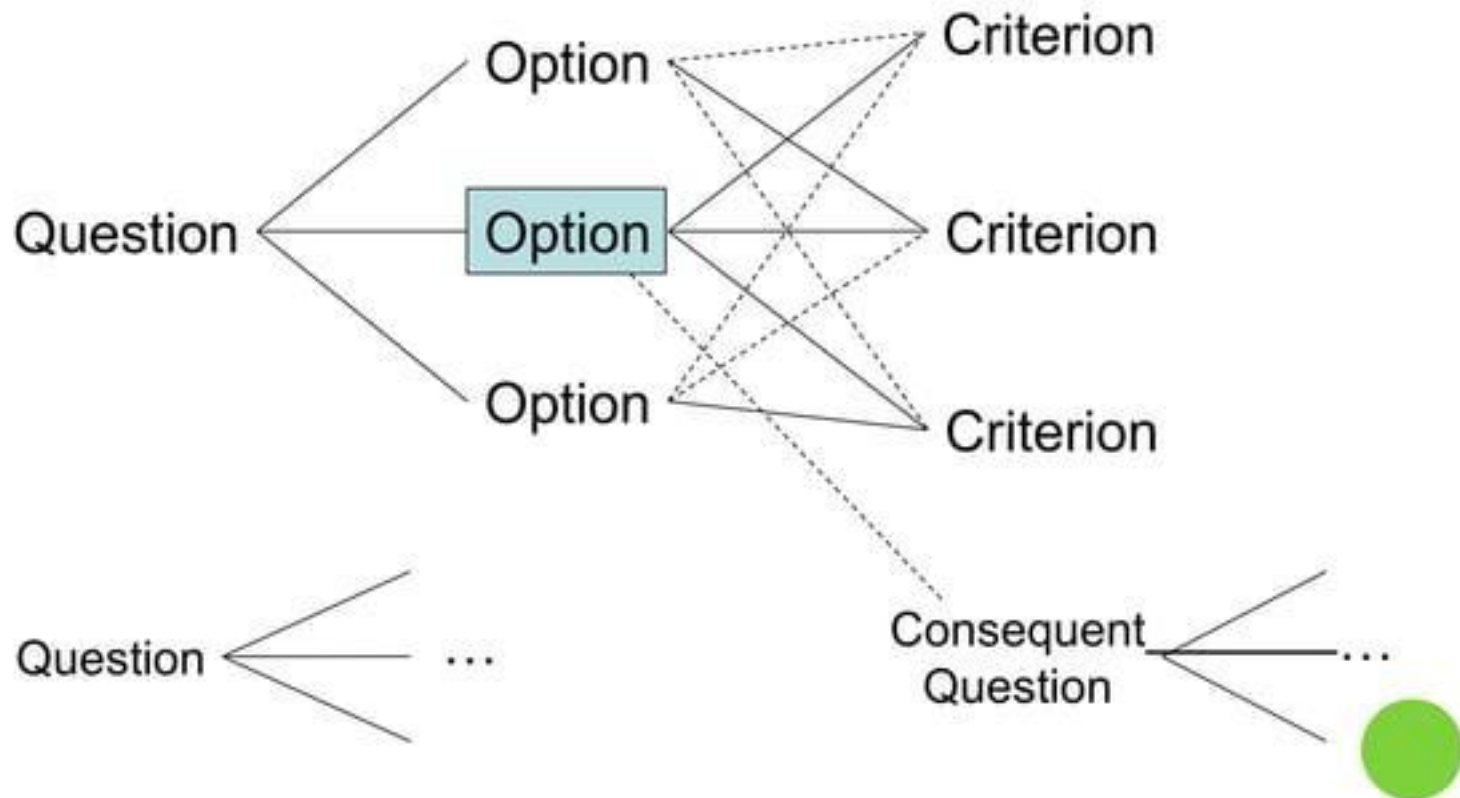
- provide alternative solutions to the question

criteria

- the means to assess the options in order to make a choice



THE QOC NOTATION



- DRL – similar to QOC with a larger language and more formal semantics
- The questions, options and criteria in DRL are given the names:
 - decision problem
 - alternatives
 - goals.
 - Advantage:
 - manage the large volume of information
 - design knowledge can be used for the design of other products.
 - Disadvantage:
 - increased overhead



PSYCHOLOGICAL DESIGN RATIONALE

- *task–artifact cycle*
 - When the new system is implemented, or becomes an artifact, further observation reveals that in addition to the required tasks it was built to support, it also supports users in tasks that the designer never intended.
 - Once designers understand these new tasks, and the associated problems that arise between them and the previously known tasks, the new task definitions can serve as requirements for future artifacts.
 - Eg: word processing



- aims to make explicit consequences of design for users
- designers identify tasks system will support
- scenarios are suggested to test task
- users are observed on system
- psychological claims of system made explicit
- negative aspects of design can be used to improve next iteration of design



DESIGN RULES

Principles
Standards
Guidelines
Rules



TYPES OF DESIGN RULES

- *Principles* are abstract design rules, with high generality and low authority.
- *Standards* are specific design rules, high in authority and limited in application
- *guidelines* tend to be lower in authority and more general in application.



PRINCIPLES TO SUPPORT USABILITY

3 main categories:

Learnability

the ease with which new users can begin effective interaction and achieve maximal performance

Flexibility

the multiplicity of ways the user and system exchange information

Robustness

the level of support provided the user in determining successful achievement and assessment of goal-directed behaviour



LEARNABILITY

Table 7.1 Summary of principles affecting learnability

Principle	Definition	Related principles
Predictability	Support for the user to determine the effect of future action based on past interaction history	Operation visibility
Synthesizability	Support for the user to assess the effect of past operations on the current state	Immediate/eventual honesty
Familiarity	The extent to which a user's knowledge and experience in other real-world or computer-based domains can be applied when interacting with a new system	Guessability, affordance
Generalizability	Support for the user to extend knowledge of specific interaction within and across applications to other similar situations	–
Consistency	Likeness in input–output behavior arising from similar situations or similar task objectives	–

PRINCIPLES OF LEARNABILITY

Predictability

- determining effect of future actions based on past interaction history
- user-centered concept;
- It is not enough for the behavior of the computer system to be determined completely from its state, as the user must be able to take advantage of the determinism.
- Eg: mathematical puzzle would be to present you with a sequence of three or more numbers and ask you what would be the next number in the sequence.
- Operation visibility* refers to how the user is shown the availability of operations that can be performed next

Synthesizability

- assessing the effect of past actions
- principle of *honesty* relates to the ability of the user interface to provide an observable and informative account of such change
- Problem: user must know to look for the change.



Familiarity

- how prior knowledge applies to new system
- Also referred as guessability; affordance.
- Eg: analogy between the word processor and a typewriter was intended to make the new technology more immediately accessible to those who had little experience with the former but a lot of experience with the latter.

Generalizability

- extending specific interaction knowledge to new situations.
- form of consistency.
- Eg: multi-windowing systems that attempt to provide cut/paste/copy operations to all applications in the same way

Consistency

- likeness in behavior arising from similar situations



PRINCIPLES OF FLEXIBILITY

Table 7.2 Summary of principles affecting flexibility

Principle	Definition	Related principles
Dialog initiative	Allowing the user freedom from artificial constraints on the input dialog imposed by the system	System/user pre-emptiveness
Multi-threading	Ability of the system to support user interaction pertaining to more than one task at a time	Concurrent vs. interleaving, multi-modality
Task migratability	The ability to pass control for the execution of a given task so that it becomes either internalized by the user or the system or shared between them	—
Substitutivity	Allowing equivalent values of input and output to be arbitrarily substituted for each other	Representation multiplicity, equal opportunity
Customizability	Modifiability of the user interface by the user or the system	Adaptivity, adaptability

PRINCIPLES OF ROBUSTNESS

Table 7.3 Summary of principles affecting robustness

Principle	Definition	Related principles
Observability	Ability of the user to evaluate the internal state of the system from its perceivable representation	Browsability, static/dynamic defaults, reachability, persistence, operation visibility
Recoverability	Ability of the user to take corrective action once an error has been recognized	Reachability, forward/backward recovery, commensurate effort
Responsiveness	How the user perceives the rate of communication with the system	Stability
Task conformance	The degree to which the system services support all of the tasks the user wishes to perform and in the way that the user understands them	Task completeness, task adequacy

PRINCIPLES OF ROBUSTNESS

Observability

- ability of user to evaluate the internal state of the system from its perceivable representation
- browsability; defaults; reachability; persistence; operation visibility

Recoverability

- ability of user to take corrective action once an error has been recognized
- reachability; forward/backward recovery; commensurate effort



PRINCIPLES OF ROBUSTNESS (CTD)

Responsiveness

- how the user perceives the rate of communication with the system
- Stability

Task conformance

- degree to which system services support all of the user's tasks
- task completeness; task adequacy



STANDARDS

- set by national or international bodies to ensure compliance by a large community of designers
 - standards require sound underlying theory and slowly changing technology
 - hardware standards more common than software
 - ISO 9241 defines usability as effectiveness, efficiency and satisfaction with which users accomplish tasks
- 

SAMPLE STANDARD

(a) A typical example of a military standard

5.1 *Subdivision of the display area*

In consideration of a simple, fast and accurate visual acquisition, the display area shall be divided into different sub-areas.

Such a division should be:

- Input area
- Output area
- Area for operational indications (such as status and alarms)



GUIDELINES

- more suggestive and general
- The basic categories of the Smith and Mosier guidelines are:
 1. Data Entry
 2. Data Display
 3. Sequence Control
 4. User Guidance
 5. Data Transmission
 6. Data Protection
- Each of these categories is further broken down into more specific subcategories which contain the particular guidelines



GUIDELINES

- abstract guidelines (principles) applicable during early life cycle activities
- detailed guidelines (style guides) applicable during later life cycle activities
- understanding justification for guidelines aids in resolving conflicts



- A major concern for all of the general guidelines is the subject of *dialog styles*, which in the context of these guidelines pertains to the means by which the user communicates input to the system, including how the system presents the communication device.

Table 7.4 Comparison of dialog styles mentioned in guidelines

Smith and Mosier [325]	Mayhew [230]
Question and answer	Question and answer
Form filling	Fill-in forms
Menu selection	Menus
Function keys	Function keys
Command language	Command language
Query language	–
Natural language	Natural language
Graphic selection	Direct manipulation

GOLDEN RULES AND HEURISTICS

- Useful check list for good design
- Different collections e.g.
 - Shneiderman's 8 Golden Rules
 - Norman's 7 Principles



SHNEIDERMAN'S 8 GOLDEN RULES

1. *Strive for consistency*
2. *Enable frequent users to use shortcuts*
3. *Offer informative feedback*
4. *Design dialogs to yield closure*
5. *Offer error prevention and simple error handling*
6. *Permit easy reversal of actions*
7. *Support internal locus of control*
8. *Reduce short-term memory load*



NORMAN'S 7 PRINCIPLES

1. *Use both knowledge in the world and knowledge in the head.*
2. *Simplify the structure of tasks.*
3. *Make things visible: bridge the gulfs of Execution and Evaluation.*
4. *Get the mappings right.*
5. *Exploit the power of constraints, both natural and artificial.*
6. *Design for error.*
7. *When all else fails, standardize.*



EVALUATION

Evaluation based on Experts

Evaluation based on Users



EVALUATION TECHNIQUES

- Evaluation tests the usability, functionality and acceptability of an interactive system.
- occurs in laboratory, field and/or in collaboration with users
- evaluates both design and implementation
- should be considered at all stages in the design life cycle



GOALS OF EVALUATION

- assess extent of system functionality
- assess effect of interface on user
- identify specific problems



EVALUATION THROUGH EXPERT ANALYSIS

- Cognitive Walkthrough
- Heuristic Evaluation
- Review-based evaluation



COGNITIVE WALKTHROUGH

Proposed by Polson *et al.*

- evaluates design on how well it supports user in learning task
- Walkthroughs require a detailed review of a sequence of actions.
- In the cognitive walkthrough, the sequence of actions refers to the steps that an interface will require a user to perform in order to accomplish some known task.
- main focus is to establish how easy a system is to learn



FOUR THINGS TO DO A WALKTHROUGH

- A specification or prototype of the system
- A description of the task the user is to perform on the system.
- A complete, written list of the actions needed to complete the task with the proposed system.
- An indication of who the users are and what kind of experience and knowledge the evaluators can assume about them.



FOR EACH ACTION, THE EVALUATORS TRY TO ANSWER FOUR QUESTIONS

- **Is the effect of the action the same as the user's goal at that point?**
- **Will users see that the action is available?**
- **Once users have found the correct action, will they know it is the one they need?**
- **After the action is taken, will users understand the feedback they get?**

to keep a record of what is good and what needs improvement in the design



HEURISTIC EVALUATION

- Proposed by Nielsen and Molich.
- A heuristic is a guideline or general principle or rule of thumb that can guide a design decision or be used to critique a decision that has already been made.
- useful for evaluating early design
- Example heuristics
 - system behaviour is predictable
 - system behaviour is consistent
 - feedback is provided



NIELSEN'S TEN HEURISTICS

1. **Visibility of system status**
2. **Match between system and the real world**
3. **User control and freedom**
4. **Consistency and standards**
5. **Error prevention**
6. **Recognition rather than recall**
7. **Flexibility and efficiency of use**
8. **Aesthetic and minimalist design**
9. **Help users recognize, diagnose and recover from errors**
10. **Help and documentation**



REVIEW-BASED EVALUATION

- Results from the literature used to support or refute parts of design.
- Care needed to ensure results are transferable to new design.
- Model-based evaluation
- Cognitive models used to filter design options
e.g. GOMS prediction of user performance.
- Design rationale can also provide useful evaluation information



EVALUATING THROUGH USER PARTICIPATION



LABORATORY STUDIES

- Advantages:
 - specialist equipment available
 - uninterrupted environment
- Disadvantages:
 - lack of context
 - difficult to observe several users cooperating
- Appropriate
 - if system location is dangerous or impractical for constrained single user systems to allow controlled manipulation of use



FIELD STUDIES

- Advantages:
 - natural environment
 - context retained (though observation may alter it)
 - longitudinal studies possible
- Disadvantages:
 - distractions
 - noise
- Appropriate
 - where context is crucial for longitudinal studies



Evaluating Implementations

Requires an artefact: simulation,
prototype, full implementation

EXPERIMENTAL EVALUATION

- controlled evaluation of specific aspects of interactive behaviour
- evaluator chooses hypothesis to be tested
- a number of experimental conditions are considered which differ only in the value of some controlled variable.
- changes in behavioural measure are attributed to different conditions



EXPERIMENTAL FACTORS

- Subjects
 - who – representative, sufficient sample
- Variables
 - things to modify and measure
- Hypothesis
 - what you'd like to show
- Experimental design
 - how you are going to do it



VARIABLES

- independent variable (IV)
characteristic changed to produce different conditions
e.g. interface style, number of menu items
- dependent variable (DV)
characteristics measured in the experiment
e.g. time taken, number of errors.



HYPOTHESIS

- prediction of outcome
 - framed in terms of IV and DV

e.g. “error rate will increase as font size decreases”

- null hypothesis:
 - states no difference between conditions
 - aim is to disprove this

e.g. null hyp. = “no change with font size”



EXPERIMENTAL DESIGN

- within groups design
 - each subject performs experiment under each condition.
 - transfer of learning possible
 - less costly and less likely to suffer from user variation.
- between groups design
 - each subject performs under only one condition
 - no transfer of learning
 - more users required
 - variation can bias results.



ANALYSIS OF DATA

- Before you start to do any statistics:
 - look at data
 - save original data
- Choice of statistical technique depends on
 - type of data
 - information required
- Type of data
 - discrete - finite number of values
 - continuous - any value



ANALYSIS - TYPES OF TEST

- parametric
 - assume normal distribution
 - robust
 - powerful
- non-parametric
 - do not assume normal distribution
 - less powerful
 - more reliable
- contingency table
 - classify data by discrete attributes
 - count number of data items in each group



ANALYSIS OF DATA (CONT.)

- What information is required?
 - is there a difference?
 - how big is the difference?
 - how accurate is the estimate?
- Parametric and non-parametric tests mainly address first of these



EXPERIMENTAL STUDIES ON GROUPS

More difficult than single-user experiments

Problems with:

- subject groups
- choice of task
- data gathering
- analysis



SUBJECT GROUPS

larger number of subjects

⇒ more expensive

longer time to 'settle down'

... even more variation!

difficult to timetable

so ... often only three or four groups



THE TASK

- must encourage cooperation
- perhaps involve multiple channels options:
 - creative task e.g. *'write a short report on ...'*
 - decision games e.g. desert survival task
 - control task e.g. ARKola bottling plant



DATA GATHERING

several video cameras
+ direct logging of application

problems:

- synchronisation
- sheer volume!

one solution:

- record from each perspective



ANALYSIS

N.B. vast variation between groups

solutions:

- within groups experiments
- micro-analysis (e.g., gaps in speech)
- anecdotal and qualitative analysis

look at interactions between group and media

controlled experiments may 'waste' resources!



FIELD STUDIES

Experiments dominated by group formation

Field studies more realistic:

distributed cognition $\Rightarrow$ work studied in context real action
is *situated action*

physical and social environment both crucial

Contrast:

psychology – controlled experiment

sociology and anthropology – open study and rich data



OBSERVATIONAL METHODS

Think Aloud

Cooperative evaluation

Protocol analysis

Automated analysis

Post-task walkthroughs



THINK ALOUD

- user asked to describe what he is doing and why, what he thinks is happening etc.
- Advantages
 - simplicity - requires little expertise
 - can provide useful insight
 - can show how system is actually use
- Disadvantages
 - subjective
 - selective
 - act of describing may alter task performance



COOPERATIVE EVALUATION

- variation on think aloud
- user collaborates in evaluation
- both user and evaluator can ask each other questions throughout
- Additional advantages
 - less constrained and easier to use
 - user is encouraged to criticize system
 - clarification possible



PROTOCOL ANALYSIS

- paper and pencil – cheap, limited to writing speed
- audio – good for think aloud, difficult to match with other protocols
- video – accurate and realistic, needs special equipment, obtrusive
- computer logging – automatic and unobtrusive, large amounts of data difficult to analyze
- user notebooks – coarse and subjective, useful insights, good for longitudinal studies
- Mixed use in practice.
- audio/video transcription difficult and requires skill.
- Some automatic support tools available



AUTOMATED ANALYSIS – EVA

- Workplace project
- Post task walkthrough
 - user reacts on action after the event
 - used to fill in intention
- Advantages
 - analyst has time to focus on relevant incidents
 - avoid excessive interruption of task
- Disadvantages
 - lack of freshness
 - may be post-hoc interpretation of events



POST-TASK WALKTHROUGHS

- transcript played back to participant for comment
 - immediately → fresh in mind
 - delayed → evaluator has time to identify questions
- useful to identify reasons for actions and alternatives considered
- necessary in cases where think aloud is not possible





Query Techniques

Interviews
Questionnaires



INTERVIEWS

- analyst questions user on one-to -one basis usually based on prepared questions
- informal, subjective and relatively cheap
- Advantages
 - can be varied to suit context
 - issues can be explored more fully
 - can elicit user views and identify unanticipated problems
- Disadvantages
 - very subjective
 - time consuming



QUESTIONNAIRES

- Set of fixed questions given to users
- Advantages
 - quick and reaches large user group
 - can be analyzed more rigorously
- Disadvantages
 - less flexible
 - less probing



QUESTIONNAIRES (CTD)

- Need careful design
 - what information is required?
 - how are answers to be analyzed?
- Styles of question
 - general
 - open-ended
 - scalar
 - multi-choice
 - ranked



UNIVERSAL DESIGN

Universal design is about designing systems so that they can be used by anyone in any circumstance



UNIVERSAL DESIGN PRINCIPLES

- equitable use - No user is excluded
- flexibility in use - adaptivity to the user's pace, precision and custom.
- simple and intuitive to use
- perceptible information - provide effective communication
- tolerance for error- minimizing the impact and damage caused by mistakes
- low physical effort – comfortable to use, minimizing physical effort and fatigue.
- size and space for approach and use – reachable and used by any user regardless of body size, posture or mobility.



MULTI-MODAL INTERACTION

- Sound in the interface
- Touch in the interface
- Handwriting recognition
- Gesture recognition



DESIGNING FOR DIVERSITY

- Designing for users with disabilities
 - *Visual impairment*
 - *Hearing impairment*
 - *Physical impairment*
 - *Speech impairment*
 - *Autism*



DESIGNING FOR DIFFERENT AGE GROUPS

- *Older people*
- *Children*

DESIGNING FOR CULTURAL DIFFERENCES



THANK YOU

